Tarun sharma

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AWS CDK: Core Concepts & Architecture

Understanding AWS Cloud Development Kit Fundamentals

What Is AWS CDK?

Define Infrastructure as Code (IaC)

AWS CDK lets you model cloud resources in code

Use Familiar Programming Languages

Supports TypeScript, JavaScript, Python, Java, and C#

Generates CloudFormation Templates

CDK code is converted to CloudFormation for deployment

Developer-Friendly Abstraction

Simplifies infrastructure creation and management

Key Building Blocks Explained

App: The Project Container

Holds one or more stacks for your infrastructure.

Stack: Deployment Unit

Maps to a CloudFormation stack; deployed as a single unit.

Construct: Reusable Building Block

Encapsulates logic and AWS resources for modular design.

Resource: Actual AWS Service

Represents services like Lambda, API Gateway, or DynamoDB.

How the CDK Workflow Operates

Write Infrastructure as Code

Use familiar languages like TypeScript, Python, Java, or C#

Synthesize to CloudFormation

CDK code generates a CloudFormation template

Provision AWS Resources

CloudFormation deploys resources from the template

Developer-Friendly Abstraction

CDK simplifies CloudFormation with higher-level constructs

Constructs: L1, L2, and L3 Levels

L1 Constructs: Direct CloudFormation Mapping

Low-level, 1:1 with CloudFormation resources (e.g., CfnFunction)

L2 Constructs: Curated with Best Practices

Higher-level, easier to use (e.g., lambda.Function)

L3 Constructs: High-Level Patterns

Combine multiple resources for common architectures

Constructs Encapsulate Logic & Resources

Reusable, modular building blocks for AWS infrastructure

Stacks and CloudFormation Internals

One-to-One Stack Mapping

Each CDK Stack directly maps to a CloudFormation stack.

Automated Resource Synthesis

CDK generates resources, IAM roles, dependencies, and outputs.

Lifecycle Management by CloudFormation

Handles create, update, and delete operations for resources.

Automatic Rollbacks on Failure

CloudFormation reverts changes if deployment fails.

Declarative Nature of CDK

Describe the Desired State

Define what you want, not how to build it

CloudFormation Handles the Process

Automatically determines create, update, and delete actions

No Manual Resource Ordering

CDK manages dependencies and sequencing for you

Key Takeaways

Simplifies Infrastructure Management

Write code to define and manage AWS resources

Built on CloudFormation

Leverages CloudFormation for reliable deployments

Core Concepts: App, Stack, Construct

Organize, deploy, and reuse infrastructure components

Declarative and Predictable

Describe desired state; CDK ensures safe changes